

Exercise I. [course]

Let f and g be two positive locally integrable functions on $[a, b]$, $b \in \overline{\mathbb{R}}$ such that $f \leq g$ on $[a, b]$. Show that :

- (8) 1. If $\int_a^b g(t) dt$ converges, then $\int_a^b f(t) dt$ converges.
(4) 2. If $\int_a^b f(t) dt$ diverges, then $\int_a^b g(t) dt$ diverges.

Exercise II.

Study the nature of the following improper integrals :

[17 marks]

- (3+18) 1. $\int_2^{+\infty} \frac{\sin(\ln x)}{x} dx$, (4) 3. $\int_1^{+\infty} \frac{\sin(x)}{x(\ln x)^{\frac{1}{2}}} dx$, (3) 5. $\int_1^{+\infty} \frac{x^3 \ln(x) dx}{e^x}$.
(3) 2. $\int_0^1 \frac{\sin(x)}{\sqrt{\cos(\frac{\pi x}{2})}} dx$, (4) 4. $\int_1^{+\infty} \left(x \ln(1 + \frac{1}{x}) - \cos(\frac{1}{\sqrt{x}}) \right) dx$.

Exercise III.

Determine whether each of the following series converges absolutely, converges conditionally, or diverges :

[12 marks]

- (2) 1. $\sum_{n=1}^{\infty} \frac{1}{\cos^2(n) + 1}$, (2+18) 3. $\sum_{n=1}^{\infty} \sin(\frac{(-1)^n}{n})$, (2) 5. $\sum_{n=1}^{\infty} \ln(1 - \frac{1}{n^2})$,
(2) 2. $\sum_{n=1}^{\infty} \frac{\ln(n)}{\arctan(n)}$, (2) 4. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n \ln(n)}$, (2) 6. $\sum_{n=1}^{\infty} \frac{(\frac{1}{2})^n e^n}{2^{\sqrt{n}} + e^n}$.

Exercise IV.

Calculate the following limits :

[8 marks]

- (6) a. $\lim_{x \rightarrow 0} \frac{1}{x^3} \int_0^x t^2 f(t) dt$, where f is a continuous function on $[-1, 1]$.
(3) b. $\lim_{x \rightarrow 0^+} \frac{1}{x} \int_{\sqrt{x}}^{\sqrt{x}} \frac{t^2}{\sin(t)} dt$.

Exercise V.

Study, according to the values of α , the nature of the following series and integrals :

[13 marks]

- (3+2+18) 1. $\int_0^{+\infty} \frac{\sin^2(x) dx}{x^\alpha + x}$, (3) 3. $\sum_{n=1}^{\infty} (e^{\frac{1}{n^\alpha}} - 1)$.
(3) 2. $\int_1^{+\infty} \frac{\cos(x^\alpha) dx}{x^{1+\alpha}}$, (4) 4. $\sum_{n=1}^{\infty} \frac{n^{2\alpha-1} + n^{\alpha+1}}{n^2 + n^\alpha}$.

Exercise VI.

[8 marks]

Let $f : [0, +\infty[\rightarrow \mathbb{R}$ be locally integrable on $[0, +\infty[$ such that $\int_0^{+\infty} f(t) e^{-t} dt$ is convergent.

- (4) 1. Let $u(x) = \int_0^x f(t) e^{-t} dt$ for all $x \geq 0$. Show that u is bounded on $[0, +\infty[$ and give $u'(x)$.
(4) 2. Show that $I(s) = \int_0^{+\infty} f(x) e^{-sx} dx$ is convergent $\forall s > 1$.
(Hint : Use part 1. and integration by parts.)

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Ex I

Given f and g be two loc int fns on $[a, b]$, $b \in \mathbb{R}$ s.t. $0 \leq f \leq g$ on $[a, b]$.

1. $\int_a^b g(t) dt$ conv $\Rightarrow \int_a^b f(t) dt$ conv??

$f(t) \leq g(t) \forall t \in [a, b] \Rightarrow F(x) = \int_a^x f(t) dt \leq G(x) = \int_a^x g(t) dt \quad \forall x \in [a, b]$.

Since $\int_a^b g(t) dt$ is convergent, then $G(x)$ is bounded & so $F(x)$ is bdd above.

Thus $\int_a^b f(t) dt$ is conv.

2. $\int_a^b f(t) dt$ div $\Rightarrow \int_a^b g(t) dt$ div?? // contr. posn of f

Suppose to contrary that $\int_a^b g(t) dt$ is conv, then by 1. we obtain

(4) $\int_a^b f(t) dt$ is conv which contradicts the fact that $\int_a^b f(t) dt$ is div. consequently $\int_a^b g(t) dt$ is div.

Ex II Study the nature of the following improper integrals.

1. $I = \int_2^{+\infty} \frac{\sin(\ln x)}{x} dx \stackrel{(1)}{=} \int_{\ln 2}^{+\infty} \sin t dt$; take $x = u(t) = e^t$ is of class C^1 and strictly increasing on $[\ln 2, +\infty[$ (Borw)
 $\stackrel{(1)}{=} \lim_{b \rightarrow +\infty} \int_{\ln 2}^b \sin t dt = \lim_{b \rightarrow +\infty} [-\cos b + \cos(\ln 2)]$ does not exist. Hence I div.

3. $J = \int_1^{+\infty} \frac{\sin x}{x(\ln x)^{1/2}} dx = \int_1^2 \frac{\sin x}{x(\ln x)^{1/2}} dx + \int_2^{+\infty} \frac{\sin x}{x(\ln x)^{1/2}} dx = J_1 + J_2$

For J_1 ; Near 1^+ : $\frac{\sin x}{x(\ln x)^{1/2}} \sim \frac{\sin 1}{1 \cdot (x-1)^{1/2}} \geq 0$ near 1^+ .

Since $\int_1^2 \frac{1}{(x-1)^{1/2}} dx$ is conv, Riemann $\alpha = 1/2 < 1$, then by equiv test J_1 is conv.

For J_2 ; Near $+\infty$: We use Abel's test.

(1) $x \cdot g(x) = \frac{1}{x(\ln x)^{1/2}}$ is cont ($\frac{\text{cont}}{\text{cont} \neq 0}$), positive on $[2, +\infty[$

(2) $\lim_{x \rightarrow +\infty} g(x) = 0$

(3) g is decreasing.

(4) $g'(x) = \frac{-\ln^{-1/2}(x) \cdot (\frac{1}{2} + \ln x)}{x^2 \ln x} < 0$ on $[2, +\infty[$

(5) $|\int_2^x \sin t dt| = |\cos x - 1| \leq |\cos x| + 1 \leq 2$.

Hence J_2 is conv.

Thus J is convergent.

③ $K = \int_1^{+\infty} \frac{x^3 \ln x}{e^x} dx$; $x^{10} \frac{x^3 \ln x}{e^x} \xrightarrow{x \rightarrow +\infty} 0$, then K is conv.

4. $L = \int_1^{+\infty} (x \ln(1 + \frac{1}{x}) - \cos \frac{1}{\sqrt{x}}) dx$;

$x \ln(1 + \frac{1}{x}) - \cos \frac{1}{\sqrt{x}} \sim x [\frac{1}{x} - \frac{1}{2x^2} + \frac{1}{3x^3}] - [1 - \frac{1}{2x} + \frac{1}{4!x^2} + \dots] \sim \frac{1}{24x^2} > 0$
 Since $\int_1^{+\infty} \frac{1}{x^2} dx$ is convergent; Riemann $\alpha = 2 > 1$, then L is convergent.

2. $M = \int_0^1 \frac{\sin(x)}{\sqrt[3]{\cos(\frac{\pi}{2}x)}} dx$; Taylor: $\cos(\frac{\pi}{2}x) \sim \cos(\frac{\pi}{2}) + (x-1)(-\frac{\pi}{2} \sin \frac{\pi}{2}) = 1 - \frac{\pi}{2}(1-x)$
 $\sim \frac{\pi}{2}(1-x) > 0$ near 1

Hence, $\frac{\sin x}{\sqrt[3]{\cos \frac{\pi}{2}x}} \sim \frac{\sin 1}{\sqrt[3]{\frac{\pi}{2}(1-x)^{1/3}}}$

Since $\int_0^1 \frac{1}{(1-x)^{1/3}} dx$ is conv, then by equiv test M is convergent.

Ex III

Determine whether each of the following series conv. abs., conv. conditionally or diverges.

1. $\sum_{n=1}^{+\infty} \frac{1}{\cos^2 n + 1}$ div, because $\lim_{n \rightarrow +\infty} \frac{1}{\cos^2 n + 1} \not\rightarrow 0$, Hence $\neq 0$.

2. $\sum_{n=1}^{+\infty} \frac{\ln n}{\arctan n}$ div, because $\lim_{n \rightarrow +\infty} \frac{\ln n}{\arctan n} = +\infty (\neq 0)$.

3. $\sum_{n=1}^{+\infty} \sin(\frac{(-1)^n}{n})$

$\sin(\frac{(-1)^n}{n}) = \frac{(-1)^n}{n} - \frac{(-1)^n}{3!n^3} + o(\frac{(-1)^n}{n^3})$

Since $\sum_{n=1}^{+\infty} \frac{(-1)^n}{n}$ is conv, Alternating Series, and $\sum_{n=1}^{+\infty} \frac{(-1)^n}{n^3}$ is conv, by comp test,

then $\sum_{n=1}^{+\infty} o(\frac{(-1)^n}{n^3})$ is conv and consequently $\sum_{n=1}^{+\infty} \sin(\frac{(-1)^n}{n})$ is conv.

Absolute conv: $|\sin(\frac{(-1)^n}{n})| \sim |\frac{(-1)^n}{n}| = \frac{1}{n}$; $\sum \frac{1}{n}$ div implies the series is not abs. conv.

4. $\sum_{n=1}^{+\infty} \frac{(-1)^n}{n \ln n}$;

① Absolute conv: $\sum_{n=2}^{+\infty} \frac{1}{n \ln n}$ div (Riemann), then the series is not abs. conv

② Convergence: Alternating series test: $\frac{1}{n \ln n} > 0, \searrow \& \xrightarrow{n \rightarrow +\infty} 0$. Hence the series converges

Ex III 5. $\sum_{n=1}^{+\infty} \ln(1 - \frac{1}{n^2})$; $\ln(1 - \frac{1}{n^2}) \sim -\frac{1}{n^2} \Rightarrow |\ln(1 - \frac{1}{n^2})| \sim \frac{1}{n^2}$

$\sum \frac{1}{n^2}$ conv; R.M., then the series conv absolutely.

6. $\sum_{n=1}^{+\infty} \frac{(\frac{1}{2})^n e^n}{2\sqrt{n} + e^n}$; $0 \leq \frac{(\frac{1}{2})^n e^n}{2\sqrt{n} + e^n} \stackrel{(1)}{\leq} (\frac{1}{2})^n$

$\sum_{n=1}^{+\infty} (\frac{1}{2})^n$ is conv, Geometric series & $|\frac{1}{2}| < 1$, then the series converges.

Since the general term is positive, then it converges absolutely.

Ex IV. Calculate the following limits

a. $\lim_{x \rightarrow 0^+} \frac{1}{x^2} \int_0^x t^2 f(t) dt$ where $f \in C([-1, 1])$.

Since,

① $x g(t) = t^2$ integrable on $[0, x]$ ($\because$ it is cont on $\mathbb{R} \supseteq [0, x]$)

① $x g(t)$ has const sign on $[0, x]$

① $x f$ is cont $[0, x] \subseteq [-1, 1]$

then by first mean value theorem, $\exists c_x \in [0, x]$ s.t

$$\lim_{x \rightarrow 0^+} \frac{1}{x^2} \int_0^x t^2 f(t) dt = \lim_{x \rightarrow 0^+} \frac{f(c_x)}{x^3} \int_0^x t^2 dt = \lim_{x \rightarrow 0^+} \frac{f(c_x)}{x^3} \frac{x^3}{3} = \lim_{x \rightarrow 0^+} \frac{1}{3} f(c_x) \stackrel{(1)}{=} \frac{1}{3} f(0)$$

($\because 0 < c_x < x \Rightarrow c_x \xrightarrow{x \rightarrow 0^+} 0$
 $\Rightarrow f(c_x) \rightarrow f(0)$
 $(\because f$ is cont at 0)

b. $\lim_{x \rightarrow 0^+} \frac{1}{x} \int_{\frac{\sqrt{x}}{2}}^{\sqrt{x}} \frac{t^2}{\sin t} dt$

Since

$x g(t) = t$ is integrable ~~on~~ on $[\frac{\sqrt{x}}{2}, \sqrt{x}]$ ($\because$ g cont on $\mathbb{R}$)

$x g(t)$ has const sign on $[\frac{\sqrt{x}}{2}, \sqrt{x}]$

① $x f(t) = \frac{t}{\sin t}$ is cont on $[\frac{\sqrt{x}}{2}, \sqrt{x}]$

then by 1st M.V.T, $\exists c_x \in [\frac{\sqrt{x}}{2}, \sqrt{x}]$ s.t

$$\lim_{x \rightarrow 0^+} \frac{1}{x} \int_{\frac{\sqrt{x}}{2}}^{\sqrt{x}} \frac{t}{\sin t} \cdot t dt = \lim_{x \rightarrow 0^+} \frac{1}{x} \frac{c_x}{\sin c_x} \int_{\frac{\sqrt{x}}{2}}^{\sqrt{x}} t dt = \lim_{x \rightarrow 0^+} \frac{c_x}{\sin c_x} \cdot \frac{1}{x} [\frac{x}{2} - \frac{(\frac{\sqrt{x}}{2})^2}{2}]$$

$$= \lim_{x \rightarrow 0^+} [\frac{1}{2} - \frac{1}{8}] \frac{c_x}{\sin c_x} \quad (\frac{\sqrt{x}}{2} \leq c_x \leq \sqrt{x} \Rightarrow c_x \xrightarrow{x \rightarrow 0^+} 0 \Rightarrow \frac{c_x}{\sin c_x} \xrightarrow{x \rightarrow 0^+} 1)$$

① $= \frac{3}{8}$

Ex VII. Study according to the values of α , the nature of the following series & inty.

$$1. I = \int_0^{+\infty} \frac{\sin^2 x}{x^\alpha + 2} dx = \int_0^1 \frac{\sin^2 x}{x^\alpha + 2} dx + \int_1^{+\infty} \frac{\sin^2 x}{x^\alpha + 2} dx = I_1 + I_2$$

For I_1 , near 0^+ : $\frac{\sin^2 x}{x^\alpha + 2} \sim \frac{x^2}{x^\alpha + 2} \sim \begin{cases} \frac{x^2}{2} = x & \text{if } \alpha > 1 \\ \frac{x^2}{x^\alpha} = \frac{1}{x^{\alpha-2}} & \text{if } \alpha < 1 \\ \frac{x^2}{2x} = \frac{1}{2}x & \text{if } \alpha = 1 \end{cases}$ (All pos. or nbd of 0)

If $\alpha > 1$, then I_1 converges (Riemann)

If $\alpha < 1$, then I_1 conv. Also because $\alpha - 2 < -1$.

Hence I_1 converges for all values of α . ①

For I_2 , near $+\infty$:

② If $\alpha > 1$, $0 \leq \frac{\sin^2 x}{x^\alpha + 2} \leq \frac{1}{x^\alpha}$. Since $\int_1^{+\infty} \frac{1}{x^\alpha} dx$ conv; Riemann, then I_2 converges.

If $\alpha = 1$: $\frac{\sin^2 x}{2x} = \frac{1 - \cos 2x}{4x}$

Since $\int_1^{+\infty} \frac{1}{4x} dx$ div (Riemann)

and $\frac{1}{4} \int_1^{+\infty} \frac{\cos 2x}{x} dx$ conv (Abel test $\frac{1}{x}$ cont, ≥ 0 , $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$ & $|\int_1^x \cos 2t dt| \leq \frac{1}{2} |\sin 2x - \sin 2| \leq \frac{2}{2} = 1$)

then by Abel test I_2 is div.

If $\alpha < 1$: $\frac{\sin^2 x}{x^\alpha + 2} \sim \frac{\sin^2 x}{x^\alpha} \gg 0$. Since $\int_1^{+\infty} \frac{\sin^2 x}{x^\alpha} dx$ div, then I_2 div.

Now, I conv iff I_1, I_2 both conv iff $\alpha > 1$.

2. $J = \int_1^{+\infty} \frac{\cos(x^\alpha)}{x^{1+\alpha}} dx$

If $\alpha > 0$: $0 \leq \left| \frac{\cos x^\alpha}{x^{1+\alpha}} \right| \leq \frac{1}{x^{1+\alpha}}$, ① $\int_1^{+\infty} \frac{1}{x^{1+\alpha}} dx$ conv $\Rightarrow J$ is absol conv & hence conv

If $\alpha = 0$: $J = \int_1^{+\infty} \frac{\cos 1}{x} dx$ div Riemann - ①

If $\alpha < 0$: $\frac{\cos(x^\alpha)}{x^{1+\alpha}} \sim \frac{1}{x^{1+\alpha}}$, $1+\alpha < 1 \Rightarrow \int_1^{+\infty} \frac{1}{x^{1+\alpha}} dx$ div (Riemann) $\Rightarrow J$ div (equiv test).

Thus J conv for $\alpha > 0$

Exv) 3. $\sum_{n=1}^{\infty} (e^{\frac{1}{n^\alpha}} - 1)$

① If $\alpha > 0$: $\alpha > 0 \Rightarrow \frac{1}{n^\alpha} \xrightarrow{n \rightarrow \infty} 0 \Rightarrow e^{\frac{1}{n^\alpha}} - 1 \sim \frac{1}{n^\alpha} > 0$

Since $\sum \frac{1}{n^\alpha}$ converges iff $\alpha > 1$, then the series conv for $\alpha > 1$ & div for $0 < \alpha \leq 1$

① If $\alpha = 0$: The general term becomes $(e-1) \not\xrightarrow{n \rightarrow \infty} 0$. Hence the series diverges

① If $\alpha < 0$: $\alpha < 0 \Rightarrow \frac{1}{n^\alpha} \xrightarrow{n \rightarrow \infty} +\infty \Rightarrow e^{\frac{1}{n^\alpha}} - 1 \xrightarrow{n \rightarrow \infty} +\infty \Rightarrow e^{\frac{1}{n^\alpha}} - 1 \not\xrightarrow{n \rightarrow \infty} 0$
 $\Rightarrow$ The series div.

4. $\sum_{n=1}^{\infty} \frac{n^{2\alpha-1} + n^{\alpha+1}}{n^2 + n^\alpha}$

$$\frac{n^{2\alpha-1} + n^{\alpha+1}}{n^2 + n^\alpha} \sim \begin{cases} \frac{n^{2\alpha-1}}{n^2} = \frac{1}{n^{2-\alpha}} & \text{If } 2\alpha-1 > \alpha+1 \Rightarrow \alpha > 2 \quad \textcircled{1} \\ \frac{n^{\alpha+1}}{n^2} = \frac{1}{n^{1-\alpha}} & \text{If } 2\alpha-1 < \alpha+1 \Rightarrow \alpha < 2 \quad \textcircled{1} \\ \frac{2n^3}{2n^2} = n = \frac{1}{n^{-1}} & \text{If } 2\alpha-1 = \alpha+1 \Rightarrow \alpha = 2 \quad \textcircled{1} \end{cases}$$

The series converges iff $\begin{cases} \alpha > 2 \text{ and } 1-\alpha > 1 \\ \text{or} \\ \alpha < 2 \text{ and } 1-\alpha > 1 \\ \text{or} \\ \alpha = 2 \end{cases}$ iff $\begin{cases} \alpha > 2 \text{ and } \alpha < 0 \\ \text{or} \\ \alpha < 2 \text{ and } \alpha < 0 \end{cases}$ iff $\frac{1}{n}$ $\alpha < 0$

Exv) Given f loc integr on $[0, +\infty[$ & $\int_0^{+\infty} f(t) e^{-t} dt$ is convergent.

1. $u(n) := \int_0^n f(t) e^{-t} dt \quad \forall n \in [0, +\infty[$

* u bdd on $[0, +\infty[$ as $\lim_{n \rightarrow +\infty} u(n) = \int_0^{+\infty} f(t) e^{-t} dt \exists$. Hence

② u is bdd near $+\infty$ for example on $[M, +\infty[$ for some $M > 0$
 but u is cont on $[0, M]$ (closed & bdd interval) implies u is bdd on $[0, M]$.
 Consequently u is bdd on $[0, +\infty[$.

② * $u'(x) = f(x) e^{-x} dx$

2. $I(s) = \int_0^{+\infty} f(x) e^{-sx} dx = \int_0^{+\infty} \frac{f(x) e^{-x}}{u'} \frac{e^{(1-s)x}}{u} dx \quad \textcircled{1}$

$= u(x) e^{(1-s)x} \Big|_0^{+\infty} - \int_0^{+\infty} u(x) (1-s) e^{(1-s)x} dx = [bdr - u(0)1] - \int_0^{+\infty} u(x) (1-s) e^{(1-s)x} dx$

Hence I & j have same nature. ~~But j is absolutely conv, then~~

But j is absolutely convergent $\because |u_n| e^{(1-\delta)x} \leq M e^{(1-\delta)x}$
 & $1-\delta < 0, x \rightarrow \infty, \frac{1}{e^{(1-\delta)x}} \rightarrow 0$